

Have you selected a topic for Assignment 3 that is different from the one that you used for Assignment 1 (i.e., have you rewritten the first three sections of the report)?

No. This report builds directly on the proposal developed in Assignment 1, with a refined focus and expanded analysis.

What feedback did your tutor provide for Assignment 1 (1%)?

[Problem Clarity]: The problem is mostly clear with some supporting detail, though the scope or goals could be sharpened.

[Business Model Analysis]: The business case is reasonable with stakeholders identified, though the value creation pathway could be developed further.

[Problem Importance]: The importance is clear and socially relevant, but the broader impact could be developed further.

[Novelty]: The proposal includes a small variation in topic or application, but the overall approach remains conventional and does not clearly establish originality.

Briefly describe how you incorporated this feedback to improve your Assignment 3 submission (maximum 150 words) (2%)

In response to the feedback, the research scope was refined from a broad comparison across healthcare and education to a focused analysis of ageing and healthcare infrastructure. This allowed for clearer problem definition and stronger alignment between research question, data, and methodology.

The business model was strengthened by explicitly linking analytical outputs (e.g., mismatch index) to decision-making use cases, such as infrastructure prioritisation and investment planning.

To address concerns about novelty, the study introduces a temporal perspective by modelling lagged demographic effects and constructing a demand–supply mismatch index. This shifts the analysis from static distribution to system responsiveness, providing a more distinctive contribution compared to existing literature.

Assignment 3: Report

Demographic Ageing and Healthcare Infrastructure in China: A Temporal Analysis of Supply Response and Regional Mismatch

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1. Introduction

China is undergoing a rapid demographic transition, with the proportion of the population aged 60 and above reaching 21.1% in 2023 (UNICEF, 2024). While the 14th Five-Year Plan (2021–2025) sets targets for expanding healthcare infrastructure, provincial responses to demographic change are often constrained by institutional and budgetary delays. Existing research primarily focuses on spatial inequality in resource distribution, but pays limited attention to the temporal dimension—specifically, how quickly infrastructure adjusts to rising demand.

This project examines the infrastructure response lag across China's 31 provinces from 2016 to 2024. Using panel data from the National Bureau of Statistics and a finite distributed lag specification, the study investigates how ageing pressure affects healthcare demand and how long it takes for supply-side capacity to respond.

By quantifying this lag, the analysis identifies provinces where infrastructure expansion fails to keep pace with demographic change. The study therefore evaluates not only the relationship between demographic structure and service provision, but also whether current resource allocation is aligned with evolving demand.

2. Related Work

Existing studies have extensively documented the spatial inequalities in the distribution of China's healthcare resources. Using Geographic Information Systems (GIS), Ma et al. (2017) reported that hospital resources are severely maldistributed across provinces, with high-quality facilities concentrated in the economically developed eastern regions. Zhang et al. (2017) quantified this inequality using Gini coefficients, revealing that while health resources have grown overall, wealthier urban populations maintain disproportionate access to hospital beds and skilled medical staff compared to those relying on primary care institutions. Most recently, Song et al. (2024) confirmed that these "spatial hotspots" of resource concentration have remained remarkably persistent over time, even at the granular county level.

While the spatial distribution of healthcare resources is well established, the temporal relationship between demographic change and infrastructure response remains less explored. Yang (2025) highlights the growing pressure of ageing and low fertility on public systems. These findings suggest delayed system responses, but do not quantify the adjustment process.

This study extends existing work by focusing on the temporal dynamics of resource allocation. By incorporating lagged demographic variables and constructing a demand–supply mismatch index, it moves beyond static analysis to evaluate how efficiently infrastructure responds to demographic change.

3. Business Model

This project sits within the domain of public sector planning and resource allocation. Its primary value lies in supporting more efficient and timely infrastructure investment in response to demographic change.

The analysis provides a diagnostic framework that identifies provinces where healthcare capacity is not aligned with demographic demand. This enables more

targeted allocation of public resources, particularly in the transition to the 15th Five-Year Plan (2026–2030).

Stakeholders and Value Creation:

- Government planners and policymakers: Identify regions with persistent under-supply, supporting prioritisation of infrastructure investment and budget allocation.
- Healthcare institutions: Understand demand–supply imbalances to inform capacity planning and expansion decisions.
- Private healthcare providers: Detect regions with unmet demand, supporting strategic investment and potential public–private partnerships.

Overall, the project provides a data-driven approach to improve the responsiveness of healthcare systems to demographic change and reduce the risk of future capacity shortages.

4. Characterising and Analysing Data:

4.1 Data Sources and Characteristics (4 V's)

Volume

The final analytical dataset contains approximately 248 observations, covering 31 Chinese mainland provinces across 8 years (2016–2024, excluding 2020), with 23 variables after cleaning and feature construction. This represents a moderate-scale dataset that can be efficiently processed in-memory using R, without requiring a distributed computing infrastructure.

Velocity

The data exhibits low velocity, as all indicators are released annually by the National Bureau of Statistics (NBS). Consequently, the project adopts a batch-processing approach, with no requirement for real-time data ingestion or streaming systems. This further supports the suitability of R as the primary analytical environment.

Variety

The dataset is structured and tabular, integrating variables from multiple domains, including demographic (e.g., Old-Age Dependency Ratio), economic (e.g., GRP per capita), and healthcare (e.g., hospital beds, outpatient visits, medical personnel per 10,000). These variables originate from different statistical yearbooks and are merged into a unified provincial panel dataset using a province–year key.

Veracity

Data is sourced from the NBS, which ensures a high level of institutional reliability. However, several limitations must be acknowledged. First, key demographic indicators such as the Old-Age Dependency Ratio (ODR) are derived from annual 1% population sample surveys rather than full censuses, introducing a small degree of sampling uncertainty. Second, some regions (e.g., Tibet) contain incomplete healthcare reporting, particularly for outpatient visits. Third, the year 2020 is entirely missing due to the COVID-19 disruption. Because this absence is systematic rather than random, imputation would distort both trend analysis and lagged variables. Therefore, 2020 is excluded from the analysis to preserve the integrity of the modelling framework.

4.2 Platforms, Software and Tools

Given the structured and moderate size of the dataset, R is sufficient for data processing, statistical modelling, and visualisation. The analysis primarily relies on the **tidyverse** ecosystem for data wrangling, particularly **dplyr** for transformation and **purrr** for handling multiple data sources.

Visualisations are generated using **ggplot2**, while spatial representations (where applicable) are supported by the **sf** package. Model diagnostics are conducted using the **car** package for Variance Inflation Factor (VIF) analysis, and the **sandwich** and **lmtest** packages are used to compute heteroscedasticity-robust standard errors (HC3). Missing data patterns are examined using the **nanian** package.

This toolset is appropriate given the dataset's scale and structure. However, if the project were extended to higher-frequency data (e.g., monthly observations) or finer spatial granularity (e.g., city-level data), in-memory processing in R may become inefficient. In such cases, a transition to a relational database system (e.g., PostgreSQL) combined with distributed processing frameworks such as Apache Spark would be more suitable.

4.3 Analytical Methods and Techniques

This project adopts a structured analytical framework that progresses from exploratory analysis to inferential modelling and finally to diagnostic evaluation.

Exploratory Data Analysis (EDA) is conducted to understand the distributional properties of variables, identify skewness, and detect missing data patterns. Correlation analysis is used to examine preliminary relationships between demographic indicators and healthcare outcomes, providing an initial assessment before formal modelling.

Multiple Linear Regression serves as the baseline modelling approach which is widely used to analyse relationships in panel data (Wooldridge, 2010). In this case, it is used to quantify the relationship between demographic structure (representing demand-side pressure) and healthcare service provision (representing supply-side capacity), while controlling for economic development through log-transformed GRP per capita. This approach enables clear interpretation of marginal effects and is appropriate for structured panel data of moderate size.

To account for delayed responses in public service systems, a **Distributed Lag Model (DLM)** is introduced as the primary extension. Distributed lag models are commonly used to capture delayed effects in time-dependent relationships (Almon, 1965; Zhu et al., 2025; Hadianfar et al., 2023). Unlike standard regression, which assumes immediate adjustment, this specification incorporates lagged values of key demographic variables (e.g., ODR at lag 1–3). This allows the model to capture how demographic pressure translates into service demand and infrastructure expansion over time.

To ensure statistical validity, **heteroscedasticity-robust standard errors (HC3)** are applied using the **sandwich** package. This adjustment is important given the cross-sectional heterogeneity across provinces, which may violate the constant variance assumption of ordinary least squares.

Additionally, a **joint F-test** is conducted to evaluate the collective significance of lagged variables. This complements individual coefficient tests and provides a more

comprehensive assessment of whether lag effects meaningfully contribute to the model.

Finally, a **Mismatch Index** is constructed based on model residuals. This index measures the deviation between observed healthcare capacity and the level predicted by demographic demand. Positive values indicate relative oversupply, while negative values indicate potential undersupply. This step extends the analysis beyond correlation, enabling the identification of regions where resource allocation may be misaligned with demographic needs.

4.4 Demonstration

The dataset is sourced from the National Bureau of Statistics (NBS) statistical yearbooks (2016–2024) and covers all 31 mainland provinces. It integrates indicators across demographic structure, economic output, healthcare capacity, and service utilisation. The dataset is publicly available at:

<https://www.stats.gov.cn/english/>

Dataset Overview

After cleaning and transformation, the final analytical dataset contains 248 observations and 23 variables, structured as a province–year panel. Key variables include the Old-Age Dependency Ratio (ODR), hospital beds per 10,000 persons, outpatient visits per 10,000 persons, medical personnel density, and log-transformed GRP per capita.

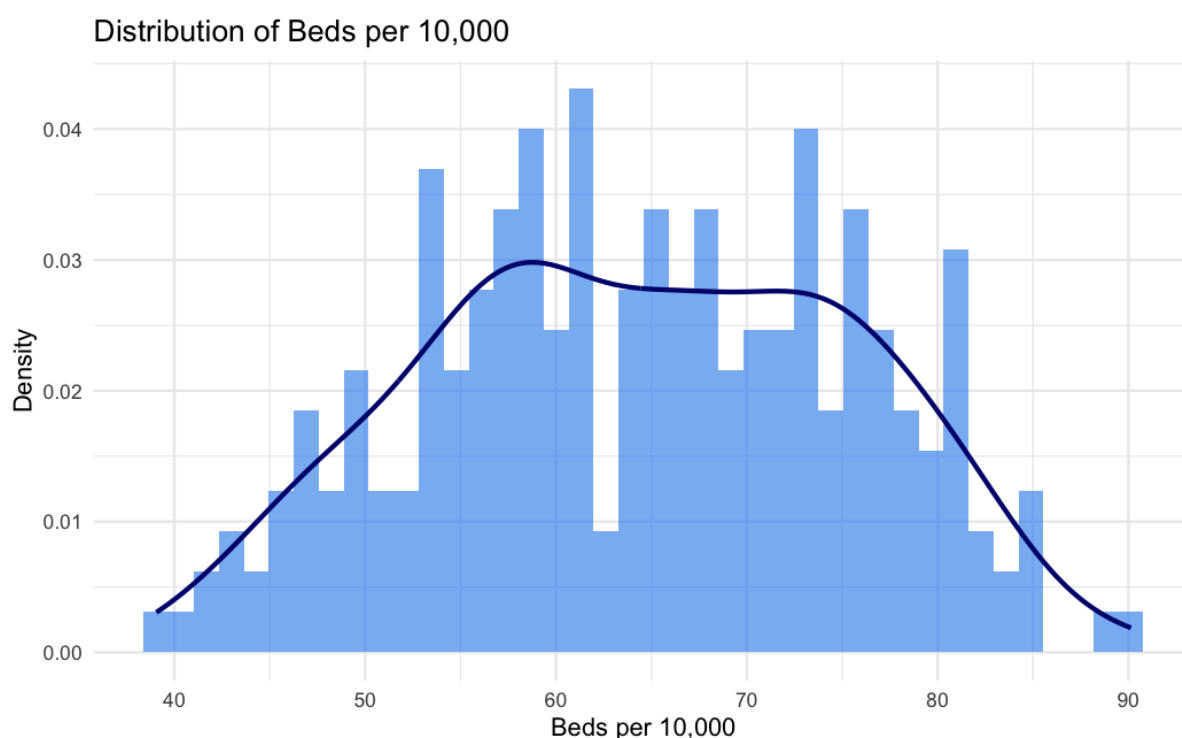


Figure 1: Distribution of beds per 10K

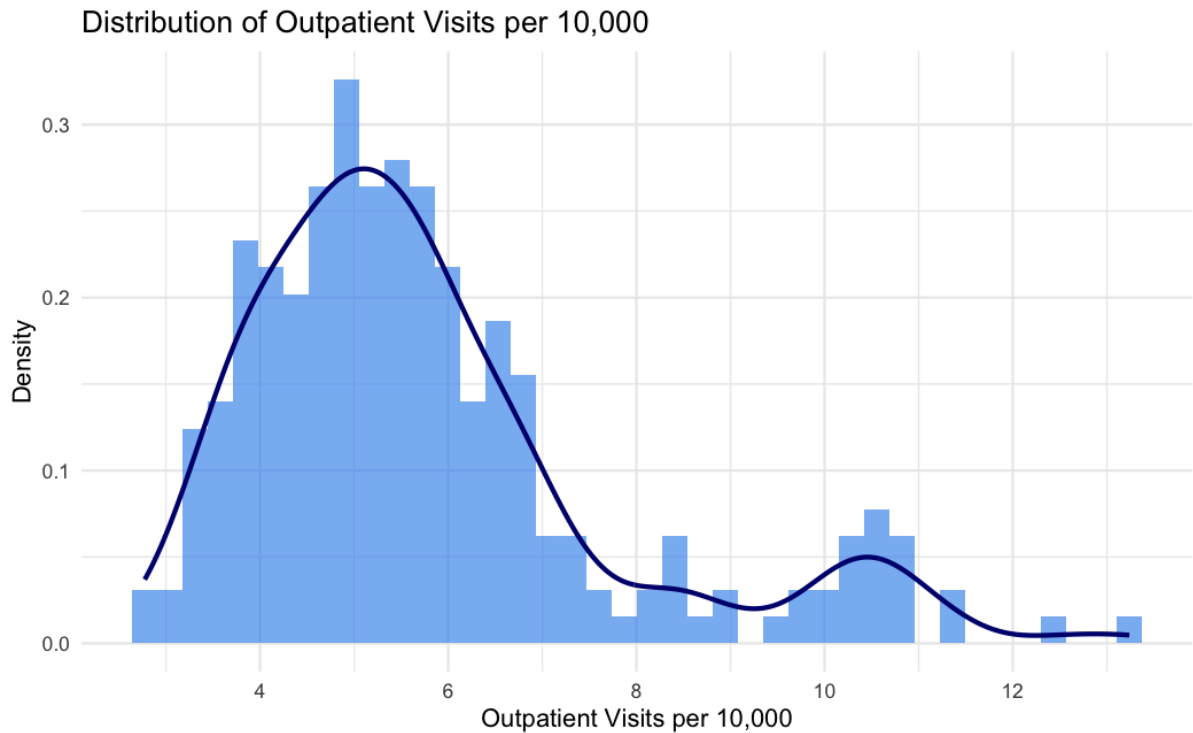


Figure 2: Distribution of outpatient service per 10K

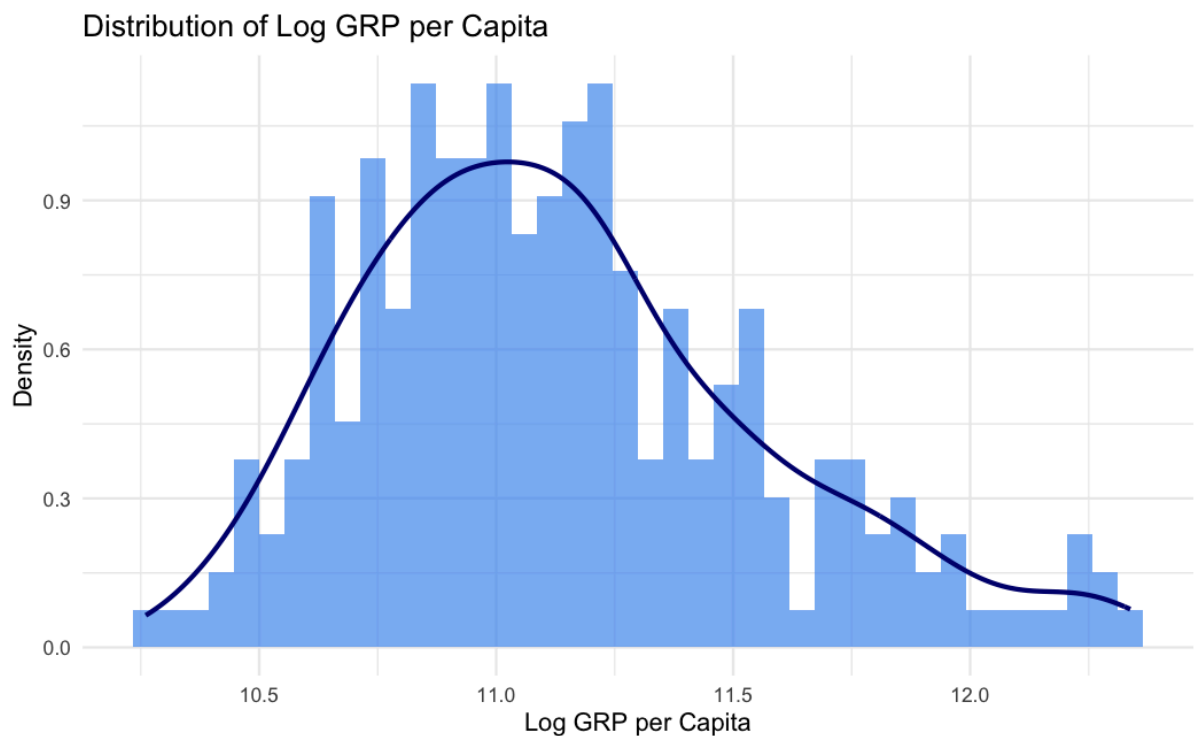


Figure 3: Distribution of log Gross Regional Product

Key distributions indicate substantial heterogeneity across provinces. Hospital bed capacity(Figure 1) shows a wide spread, reflecting uneven infrastructure provision. Outpatient visits are right-skewed(Figure 2), with a small number of provinces exhibiting very high utilisation levels. GRP per capita is log-transformed to reduce skewness(Figure 3) and improve interpretability in regression models.

Basic Characteristics

Summary statistics reveal substantial heterogeneity across provinces. The Old-Age Dependency Ratio (ODR) ranges from approximately 7% in younger western regions such as Tibet to over 30% in heavily aged northeastern provinces such as Liaoning. Similarly, hospital bed density varies widely, indicating uneven distribution of healthcare infrastructure. These patterns are consistent with prior literature on regional inequality and provide a strong empirical basis for further modelling.

Correlation Matrix: Healthcare & Demographic Variables

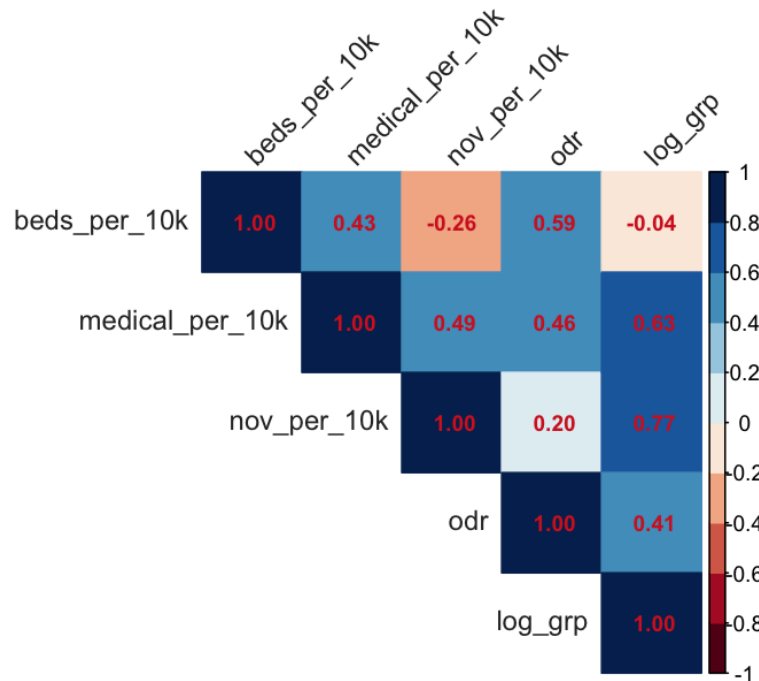


Figure 4: Correlation matrix

The correlation matrix (Figure 4) indicates a positive association between lagged ODR and outpatient visits, offering preliminary support for the delayed demand hypothesis. Economic development (log GRP per capita) is also positively correlated with both demand and supply indicators, supporting its role as a control variable in subsequent models.

Model A Results – Demand Side

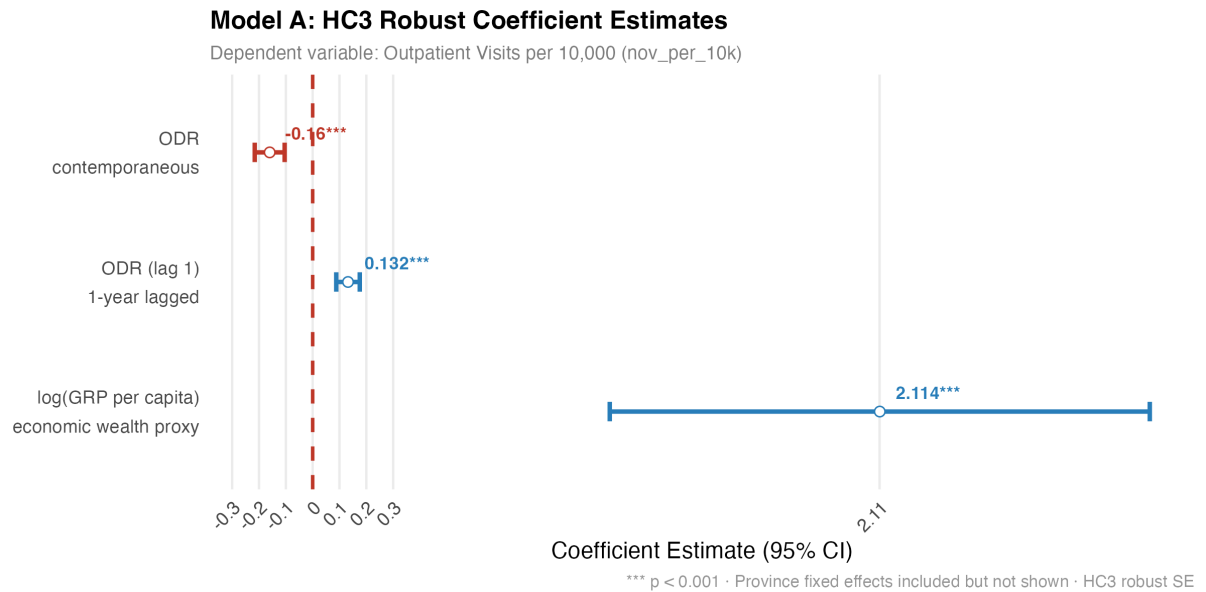


Figure 5: Coefficient Estimate of Model A

Model A examines outpatient visits per 10,000 as a function of contemporaneous and one-year lagged ODR, controlling for economic capacity (log GRP per capita) and province fixed effects. The results reveal a clear temporal pattern: the contemporaneous ODR coefficient is negative ($\beta = -0.160$, $p < 0.001$), while the lagged effect is positive ($\beta = +0.132$, $p < 0.001$). This suggests that ageing pressure does not immediately translate into increased healthcare utilisation; instead, demand rises with a delay. Economic capacity remains the strongest predictor ($\beta = +2.113$, $p < 0.001$), reflecting higher service utilisation in more developed regions.

Model B Results — Supply Side and Mismatch Index

Model B adopts a finite distributed lag specification, incorporating up to three years of lagged ODR to explain hospital bed density. A joint F-test confirms that these lagged terms are collectively significant ($p < 0.05$), indicating that infrastructure adjustment occurs over multiple years rather than instantaneously.

A mismatch index is constructed to assess whether infrastructure provision aligns with demographic demand. It is defined as the difference between observed bed capacity and the level predicted by demographic and economic factors. Provinces with persistently negative values are interpreted as under-supplied, while positive values indicate relative over-supply.

Provincial Healthcare Mismatch Index

Red = Demand > Supply; Blue = Demand < Supply

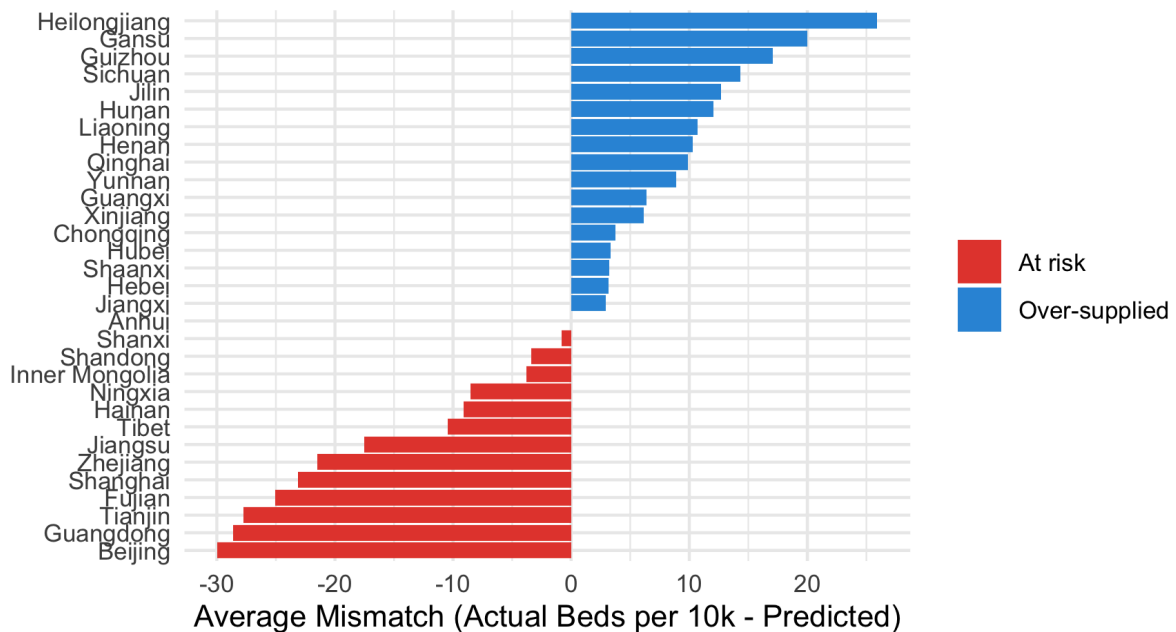


Figure 6: Mismatch index bar chart

The bar chart shows the mismatch index, defined as the difference between actual and predicted bed capacity. Provinces at high risk are concentrated in economically developed coastal regions, including Beijing, Guangdong, Tianjin, Shanghai, and Zhejiang. In contrast, several inland and northeastern provinces, such as Heilongjiang, Gansu, and Guizhou, exhibit relative oversupply.

Temporal trends provide further insight into system adjustment. For example, Shanxi shows a rapidly widening negative gap, indicating increasing pressure, while Beijing's gap is widening at a slower rate. In contrast, Guangxi and Henan show improving alignment, with infrastructure expansion outpacing demographic pressure.

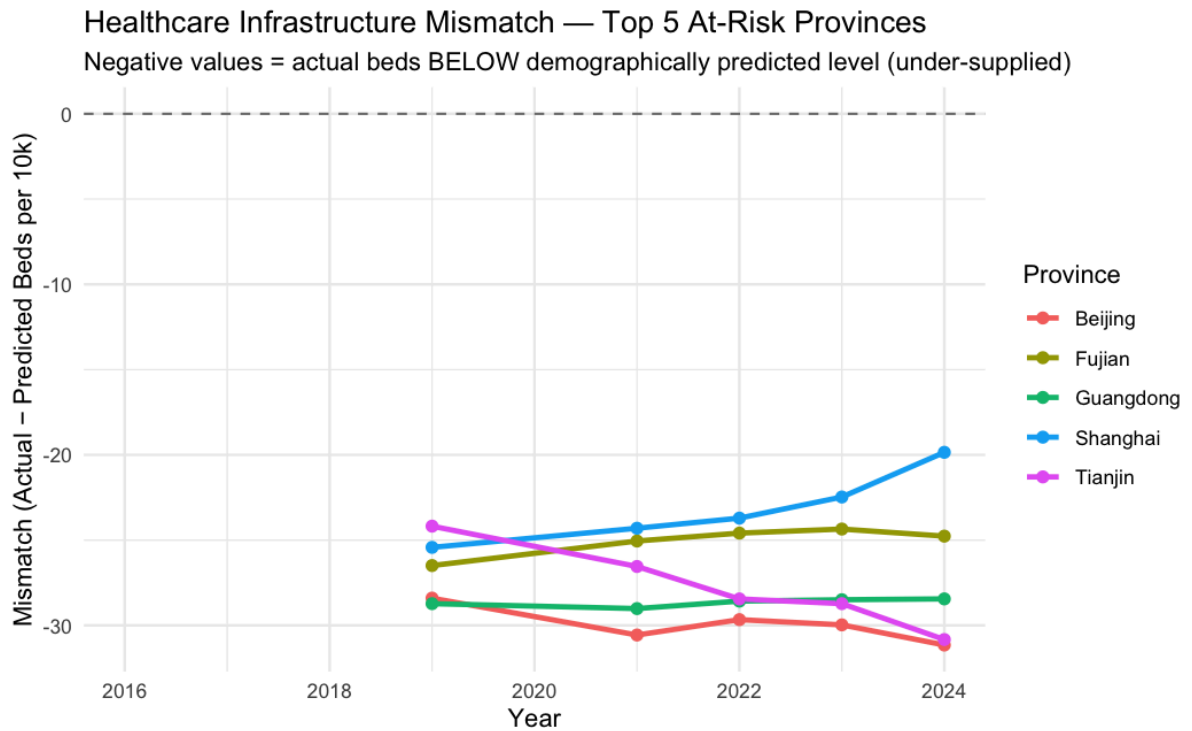


Figure 7: Mismatch index of the top 5 high-risk provinces

The mismatch analysis covers 2019–2024 (excluding 2020), as the three-year distributed lag structure requires ODR observations from 2016 onwards to compute lag terms.

Shanghai shows a significant upward trend, indicating aggressive infrastructure investment that is successfully closing the gap between demographic demand and physical capacity. Conversely, Beijing and Tianjin exhibit widening negative gaps.

These patterns are consistent with the well-documented phenomenon of inter-provincial patient migration in China, where major urban centres function as regional healthcare hubs. As a result, local demographic indicators may underestimate actual service demand, contributing to persistent undersupply in these regions.

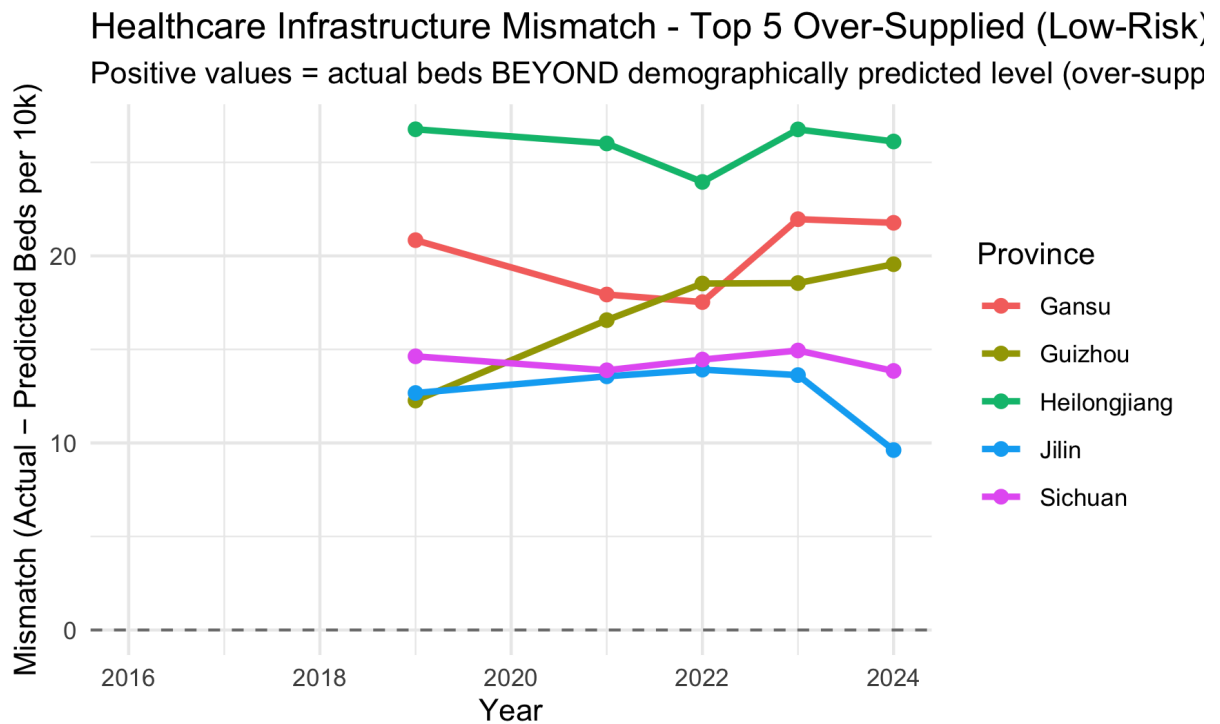


Figure 8: Mismatch index of the top 5 low-risk provinces

The figure highlights provinces with a positive mismatch, where actual bed capacity exceeds demographic benchmarks.

Provinces with positive mismatch values, such as Heilongjiang and Gansu, appear to have infrastructure levels exceeding what demographic structure alone would predict. This may reflect historical investment patterns or population outflows, which reduce local demand. While these regions face a lower immediate risk of capacity shortages, declining trends in some cases (e.g., Jilin) suggest that ageing pressure may gradually erode this surplus.

This confirms that infrastructure adjustment is not only delayed but unevenly distributed, with demographic pressure concentrating in provinces least able to absorb it through existing capacity.

5. Standard for Data Science Process, Data Governance, and Management

This project follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework (Hotz, 2022) to ensure a structured and reproducible workflow. The process begins with business understanding, where the key problem is defined as the misalignment between demographic demand and healthcare infrastructure. In the data understanding and preparation stages, data from the National Bureau of Statistics (NBS) is cleaned, transformed into panel format, and enriched with lagged variables to capture temporal effects. The modelling stage applies regression and lag analysis to quantify both immediate and delayed responses, while the evaluation stage assesses statistical validity and interprets results through the mismatch index. Finally, the deployment stage translates findings into policy-relevant insights for resource allocation.

From a data governance perspective, the dataset is publicly available and aggregated at the provincial level, which avoids issues related to personal data privacy. All data is stored and processed locally in R, ensuring controlled access and reproducibility through scripted workflows. No sensitive or confidential information is involved.

The mismatch index should be interpreted as a planning tool, not a performance judgement. Provinces showing relative over-supply – such as Heilongjiang or Gansu – should not face funding cuts on that basis alone, as surplus bed capacity in less-populated or geographically remote regions may serve genuine access needs that raw density figures do not capture. Similarly, findings should not be used to stigmatise specific provinces, but to inform a more equitable national allocation strategy.

6. Reference

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